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TEP4854 – Plastic Free Oceans

Supporting Document – Team 3

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Declaration on use of artificial intelligence tools

In working on this assignment, we used artificial intelligence (AI) tools in the following ways: we used OpenAI *ChatGPT*, Anthropic *Claude* and Mistral AI SAS *Mistral 7B* to assist with idea development and structuring of the text. AI was furthermore used to suggest phrasing and provide language improvements, but all content has been reviewed, edited and academically assessed by us. AI was not used to generate entire paragraphs or to answer the assignment in its entirety. We have also not uploaded personal data or sensitive information into the AI tool. We are solely responsible for the academic content of the assignment.

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1 Project introduction

The goal of the *Experts in Teamwork* course was to understand team dynamics and apply methods that improve collaboration and efficiency in a real project setting. The course included facilitated teamwork sessions, reflections, and analytical tasks designed to strengthen interdisciplinary cooperation. The final outcome consisted of a process report, poster design, supported by this document, and a presentation at the end of the course.

Within the theme *Plastic Free Oceans*, our team decided to develop a solution that could realistically address the growing global plastic pollution problem. To approach this challenge, we conducted literature research to determine what defines a "perfect" household HDPE packaging bottle. One that is safe, functional, and optimized for circular use. Our design concept aligns with the EU Packaging and Packaging Waste Directive (PPWR) and follows RecyClass design-for-recycling guidelines, ensuring compatibility with existing recycling systems in the European and Norwegian markets. [1][2]

The perfect household packaging bottle is not just a container, but a carefully engineered system that balances safety, sustainability, and circularity throughout its entire life cycle, from consumer use to reintegration into the recycling stream as new material.

Our concept builds on HDPE, a widely recyclable polymer that can be reused more than ten times when it is unpolluted and meets the structural requirements. By using compatible materials for the body, cap, and label, the design avoids barriers that hinder sorting and recycling. Structural optimization reduces material thickness while maintaining durability, lowering resource consumption and transport-related CO_2 emissions.

Safety and usability are ensured through child-resistant closures that remain accessible for adults and seniors. A light color strategy enables reliable detection by infrared sorting systems, improving recycling capture rates. At the same time, brand identity can still be expressed through shape, texture, and label design without compromising recyclability. [2]

The result is a packaging concept designed for continuous circulation within a European recycling system. A bottle intended to be collected, recycled again and again.

References

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